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Self-organizing Services for Browsing the World: Challenges and Directions

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Motivations

- Computer-based systems and sensors will be soon embedded in everywhere
  - all our everyday objects
  - all our everyday environments
- Current deployments of pervasive services and sensor networks focus on special purpose systems, e.g.,
  - E.g., environmental monitoring and healthcare
- General purpose approaches are likely to emerge soon
  - Shared infrastructures of sensors, tags, cameras, Web 2.0 data
  - Collecting general purpose data about physical and social world
  - Defining a sort of distributed digital "***world model***"
  - For general-purpose exploitation by "**browsing the world**" services



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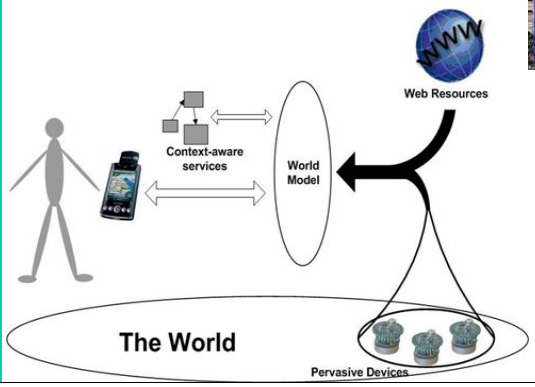
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"Browsing the World" Scenario

Exploit the world model to

- Implement services to help us interact with the physical world (e.g., personalized real-time maps)
- Have services coordinate with each other in a context-aware fashion to achieve global goals (e.g., traffic control systems)





PLEASE NOTE: Users and services are themselves part of the world, and thus must be part of the world model



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Why Self-organization?

- The complexity, openness, and dynamics of the scenario makes it impossible for humans to stay in the control loop
  - Services must autonomously organize their activities and must self-adapt to the current situation of the world
  - Humans can intervene only on limited portions of the scenario
- But…..is the challenge really with services?



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Mechanisms of Self-organization

- Direct interactions
  - Components (services/agents) interact directly with each other to reach common goals in a self-organizing way
  - Not suitable to open and dynamic systems (who are the others?)
  - Not suitable to situated activities (where most issues relates to what's happening in the world, rather than to what the others are doing)
- Mediated interactions
  - Self-organization takes place via sensing/affecting a common environment (stigmergic interactions, as in ant colonies)
  - The environment has its own properties and processes which rules the behavior of the colony (diffusion and evaporation of pheromones)
  - Components do not need to know each other (ants are blind)
  - Self-organization is by definition situated
- In the latter case, then, the real challenge relates to **engineering the environment** rather than services
  - In our case, **engineering the world model**!!



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Engineering the World Model

- Extreme heterogeneity of data
  - Information coming from sensors, cameras, Web 2.0, etc.
  - Variable density of devices (potentially continuum)
- Inherent dynamism and decentralization
  - Devices (and the associated data) come and go at any time
  - No way to control each devices due to decentralization
- Massive amounts of data produced
  - No way to collect all data (potentially infinite) at a place
  - Need to aggregate, prune, evaporate, analyse data where it is produced
- Uncertainty and reliability
  - No user/services can be ensured the availability of specific data
  - Still services must be able to go on in any case
- The necessity arises to **exploit self-organization within the world model**!!!
  - Conceptual shift from self-organizing services to self-organizing data



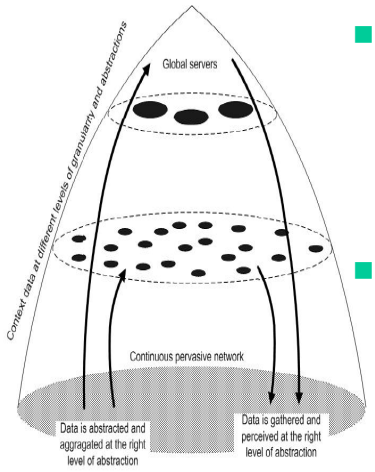
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The Knowledge Pyramid

Data becomes **knowledge** from the services' viewpoint (something the services know about the world…)



- Data virtually generated in both
  - A sensor network continuum
  - A level of Web 2.0 services
  - At any level in between (e.g., cameras, local servers, etc.)
- Data should flow up the pyramid
  - In aggregated forms, via proper self-organizing algorithms for data aggregation
  - To limit the amount of data (potentially infinite) to be managed by services
- Data should flow down the pyramid
  - To self-aggregate data coming from low-level sensors with data coming from higher-level ones (or from the Web)
  - To let agents exploit all available data in a uniform way



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Our Current Research

- Study and experience general abstractions, self-organization algorithms and associated infrastructures to engineer, via self-organization, the world model, so that it can act as a mean way to deploy self-organizing browsing the world services

- In particular:
  1. Algorithms to self-aggregate data to facilitate their exploitation by services → **self-organizing spatial regions**
  2. Identify a simple yet general-purpose mean to represent context knowledge, information, at whatever level it is produced, so as to build an effective and usable world model → **the W4 model**
  3. Define a general situated agent model and the associated infrastructure for autonomic communication services for browsing the world → **the browsing the world infrastructure**



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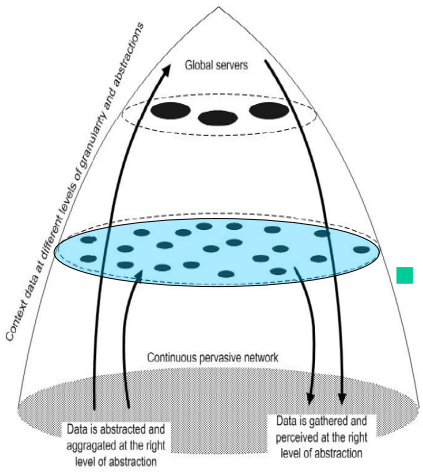


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Self-organizing Spatial Regions



- We need to aggregate data from the continuum
  - Compact representation, manageable by services
  - Without losing relevant information
  - And indeed provide more information of "what's happening"
- Key idea
  - Identify discrete aggregation regions
  - Enable "per region" views of specific characteristics of the environment



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Identifying Aggregation Regions

- Our "Region Aggregation Noise" (RAN) approach, considers the following:
  - A distributed algorithm is continuously running in the network as a sort of *"background noise"*
  - with the goal of partitioning the sensor network into regions characterized by similar patterns (as an overlay of virtual links)
  - Abstracting from the specific density/structure of the sensor network

```
Do_forerever:
      Wait(t);
      neigh[] = Select_neighbor(num_neigh);
      Foreach(neigh[])
            Data = Exchange_data();
            Update_link(data);
Done
```

```
Update_link:
if D(v(si),  v(sj)) < T {
      l(si,sj) = min(l(si,sj) + delta, 1)
} else {
      l(si,sj) = max(l(si,sj) - delta, 0)
}
```



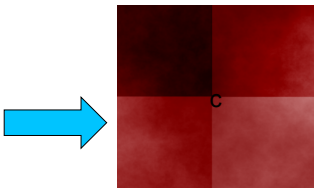
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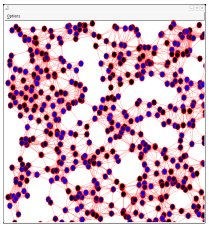
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Simulated Region Formation

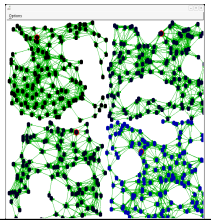
- Simulated sensors are embedded in an environment characterized by different patterns of sensed data



- At first they are not logically connected with each other



- Gradually they recognize regions
- Then a partitioning emerges based on the environmental patterns





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Aggregation on a "per region" Basis

- Identification of region is, per se, aggregated info about some characteristic of the environment
  - But we may also with to gather specific aggregated information
  - i.e., computing specific functions over a region
- Our "macro-programming" solution
  - Enable injecting of specific local self-org aggregation rules in devices (no matter how much and how dense they are)
  - Have these rules execute within the same basic aggregation scheme, at little or no additional costs
  - Have aggregated data be computed



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Macro Programming Regions

- Once regions are formed,
  - *Aggregation of sensed data* can occur on a per-region basis
  - Simply by injecting a self-organizing (gossip-based) aggregation function on the network
  - That exploit the existing aggregation noise without incurring in additional communication costs
  - Users/services can, on need, be provided with such aggregated data representing some "macro" property of a region at very limited costs
- Example: Minimum Value in a Region

```
Do_forerever:
  Wait(t);
  neigh[] = Select_neighbor(num_neigh);
  Foreach(neigh[])
        Data = Exchange_data();
        Update_link(data);
        if(connected) Local_aggregation();
Done
```

```
Local_aggregation:
if(localMini>localMinj)
        localMini=localMinj
```



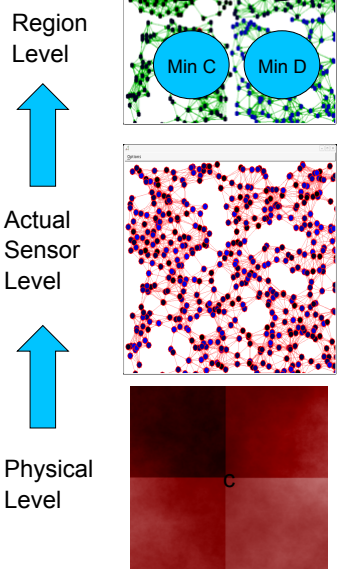
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And Eventually…

- As in the general figure of the knowledge pyramid…
- We obtain a sort of "knowledge network" describing some characteristic of the environment from different viewpoints (abstract regions level)
  - Working in a decentralized, self-organizing, self-adaptive way
  - Dealing with variable (potentially continuum) sensor densities





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Data Representation

- Once we have algorithms and tools to gather information from the world
  - And we may have huge amount of data pieces representing different aspects of the environment
- How can we provide a uniform representation of data
  - Coming from different devices
  - Expressing different levels of observations
  - Expressing different facts about the world
- Key guidelines
  - Keep it simple
  - Keep it intuitive
  - Keep it computable
- How do we usually characterize facts about the world?



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The W4 Context Model

- A simple model in which context data is expressed by a four field structure: *Who*, *What*, *Where* and *When*.
- Someone or something (*Who*) does some activity (*What*) in a certain place (*Where*) at a specific time (*When*)
- Who is acting? What is he/she/it doing? Where and when the action takes place?



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Who, What, Where, When

- **Who** is the subject. It is represented by a string with an associated namespace that defines the "kind" of entity that is represented.
  - "person:Gabriella", "tag:tag#567", "sensor-region:21"
- **What** is the activity performed. It is represented as a string containing a predicate-complement statement.
  - "read:book", "work:pervasive computing group", "read:temperature=23".
- **Where** is the location to which the context relates.
  - (longitude, latitude),
  - "campus", "here"
- **When** is the time duration to which the context relates
  - 2006/07/19:09.00am - 2006/07/19:10.00am
  - "now", "today", "yesterday", "before"



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W4 Atoms

- Gabriella is walking in the campus' park. An agent running on her PDA can periodically create an atom describing her situation.

Who: user:Gabriella
What: works:pervasive computing group
Where: lonY, latX
When: now

- Gabriella's PDA is connected with a RFID tag reader. A specific RFID agent controls the reader and handles the associated events

Who: tag:statue of Ludovico Ariosto
What: -
Where: lonY, latX
When: now



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Simple W4 Queries

- Gabriella is walking in the campus, and wants to know if some colleague is near. She will ask (read operation):

Who: user:*
What: works:pervasive computing group
Where: circle,center(lonY,latX),radius:500m
When: now

- Analogously, Gabriella can ask if some of her colleagues has gone to work in the morning:

Who: user:*
What: works:pervasive computing group
Where: office
When: 2006/07/19:09.00am - 2006/07/19:10.00am



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W4 Self-organized Knowledge Networks

- W4 Atoms can be related to each other
  - To represent relations between pieces of data and enable navigating related facts of the world
  - To refer to individual sources of aggregated data and enable vertical navigation (shifting of observation level)
- Can we exploit self-organization approaches?
  - Self-aggregation of data via a semantic extension of the spatial region approach (cluster and link related data)
  - Have data diffuse and evaporate the same as pheromone does in ant-based systems
- Still to be investigated!



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What is a Situated Service for Browsing the World?

- Browsing the world services
  - Location-dependent queries, Social interactions, Real-time understanding of situations, etc.
- Autonomic communication services
  - Distributed cooperation, Ad-hoc communications, etc.

- Given the availability of W4 spaces and data (we assume that there are a multiplicitly of accessible “W4 space” where to store locally produced W4 tuples),
  - we have to code specific software components that can somewhat “query” the W4 tuple spaces for
  - Achieving context-awareness and adapt to the current situation
  - Navigating the world model and extract high-level information about situation occurring in the environment
  - In autonomy
- So it is a **software agent**



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Key Features of Services/Agents

- We require services the capability of
  - querying the W4 space for extracting info
  - injecting new knowledge atoms and/or new aggregation algorithms in the space

```
KnowledgeAtom[ ] read(KnowledgeAtom a);
void inject(KnowledgeAtom a, Behavior b);
```

- In addition
  - Services themselves will be represented by some sorts of atoms in the space
  - Services can thus indirectly interact via the W4 space, which is a sort of



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Browsing the World Services vs. Ants

- Services produce and read W4 tuples
  - The same as ants release and sense pheromones
- Services indirectly interact via the W4 distributed environment
  - Stigmergy!
- The environment describe something about the environment
  - "There is another agent near here"…the same as pheromones do
- The environment is active
  - Aggregation and pruning of W4 atoms → similar to diffusion and evaporation of pheromones
- We still have to fully unfold these issues, and define a simple and usable agent model…
  - Yet, we have a prototype infrastructure…

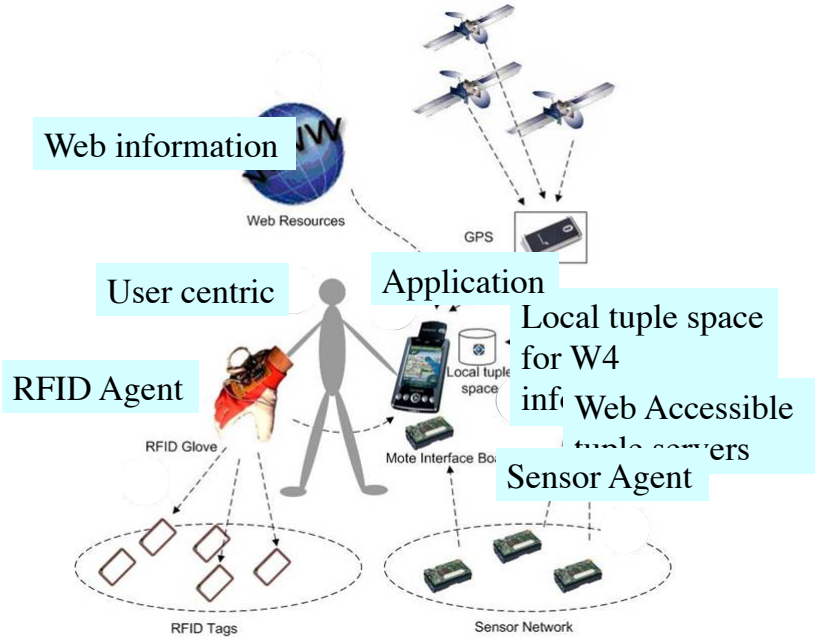


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Browsing the World Architecture





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Implementation

- The whole system has been realized using the Java language.
- **Web (global accessible) Tuple Spaces** has been implemented through a Postgres database with spatial extensions.
- The **Local Tuple Space** is simply implemented by a Java Vector.
- The RFID reader and the Mote Sensors are accessed via JNI and TCP/IP.
- The **"W4 Query Engine"** interrogates the web accessible tuple space through Sql and its postgis spatial extensions. Local tuple space instead makes use of String parsin and java algorithm.
- User interface is provided by:
  - Google Earth (for laptops) and Google Maps accessed via the Minimo browser (for PDAs).
  - Google KML Language
  - Jsp e JavaScript



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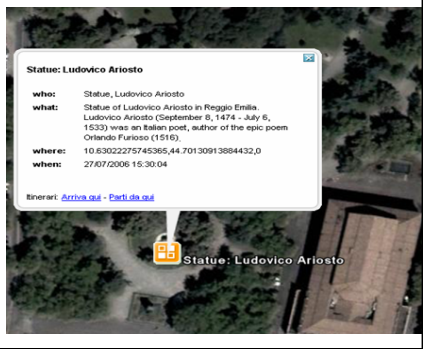
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Application: The Journey Map

- A tourist wants to automatically build and maintain a diary of his journey:
  - track of all the user movements
  - access available tourist information stored in RFID tags attached to monuments and art-pieces

- **Who:** rfid:*
- **What:** *
- **Where:** *
- **When:** now







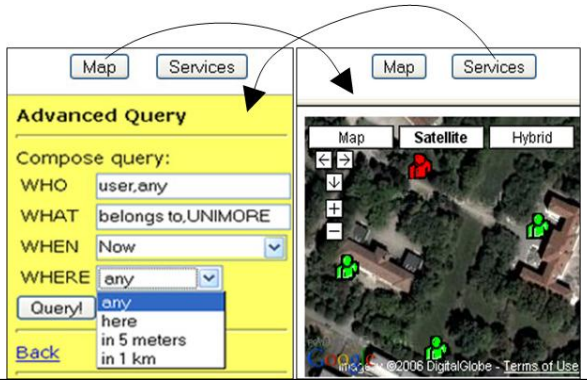
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Application: Real Time Maps

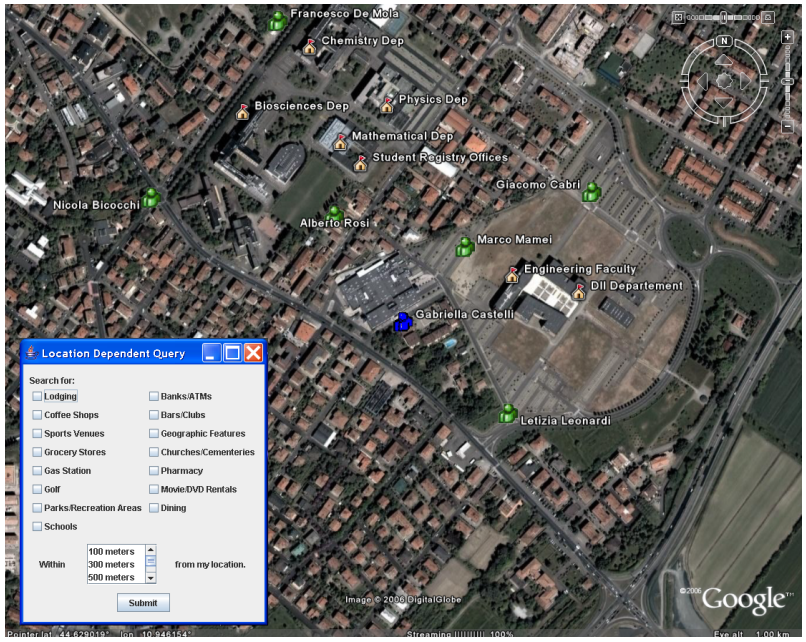
- Friends/members of a rescue time can share their actual GPS locations (W4 knowledge atoms)
  - They can be used to see each other
  - And to coordinate their movements with each other and with the environment



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Application: Real Time Maps



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Conclusions and Future Work

- Future pervasive computing scenarios invites considering
  - A continuum knowledge pyramid for pervasive networks
  - Self-organization of knowledge → Engineering the Environment
  - Autonomic services for browsing the world
- Beside our preliminary and incomplete proposals, there is indeed need to study
  - Self-organizing algorithms for knowledge aggregation
  - Proper models for representing, accessing, and integrating heterogeneous contextual data
  - Suitable “ant-like” service models exploiting the above for the provisioning of autonomic communication services